



Arab Academy for Science & Technology & Maritime Transport College of Engineering
and Technology
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Department of Computer Engineering

Course Title: Microprocessor Systems

Smart Band for Comprehensive Health Monitoring and Data Analysis

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Team Contributions Statement

All team members – Mohammed Mustafa Kamel, Abdualloh Mohamed Hussein, and Moustafa Ahmed Mahmoud – contributed collaboratively and equally across all phases and aspects of this project. This includes, but is not limited to, conceptualization, hardware selection and assembly of the **maquette**, embedded systems programming for the ESP32, sensor integration and calibration (MPU6050, MAX30105), implementation of data transmission protocols (Wi-Fi, MQTT), backend server development (Flask), data storage mechanisms (CSV-based), frontend web interface design (HTML, CSS, JavaScript, Chart.js), user authentication logic, system testing, data analysis, and the compilation of this technical report. The project was a unified team effort, with shared responsibilities in problem-solving and decision-making throughout the development lifecycle.

Abstract

This project details the design, development, and implementation of a smart band for real-time health monitoring and data analysis. The system focuses on collecting vital physiological data—including heart rate, blood oxygen saturation (SpO₂), body temperature, and step count—utilizing an ESP32 microcontroller interfaced with MAX30105 and MPU6050 sensors integrated into a physical **maquette**. Data is transmitted wirelessly via Wi-Fi and MQTT protocol to a central server, where it is stored and made accessible to users through a web-based dashboard featuring visualizations. The primary goals were to create a functional prototype **maquette** capable of active experimentation and data collection, implement a robust data pipeline from sensor to server, and provide meaningful **data analysis** and interpretation for users, addressing a real-world need for accessible personal health monitoring.

1. Introduction

The proliferation of wearable technology has revolutionized personal health management, offering individuals unprecedented access to their physiological data. This project is motivated by the increasing demand for affordable and comprehensive health monitoring solutions that empower users to track their well-being proactively. Traditional healthcare can be reactive and episodic; continuous monitoring via smart wearables provides a stream of data that can indicate trends, potential health issues, or the efficacy of lifestyle changes.

The main contributions of this work are:

1. The design and implementation of a functional smart band **maquette** prototype capable of collecting and displaying heart rate, SpO₂, temperature, and step count.
2. Development of a robust data pipeline leveraging **Wi-Fi** and **MQTT** for real-time data transmission from the wearable device to and from a **remote server**.
3. Creation of a **Flask-based web application** for user authentication, data storage (**CSV-based**), and **visualization** of health data through interactive charts.
4. Demonstration of active experimentation with a microprocessor system, including sensor data acquisition, **data analysis**, and interpretation.

This report details the system architecture, hardware and software components, data handling methodologies, and the results obtained, aligning with the project's aim to address a real-world problem through applied microprocessor system design.

2. Background

The domain of this project is personal health and wellness monitoring through wearable technology. The key physiological parameters tracked by our smart band are:

- **Heart Rate (HR):** Measured in beats per minute (BPM), heart rate is a critical indicator of cardiovascular health. Abnormalities or trends in HR can signify various conditions or stress levels.
- **Blood Oxygen Saturation (SpO₂):** This measures the percentage of oxygenated hemoglobin in the blood. Low SpO₂ levels can indicate respiratory or circulatory problems. Values are typically expected to be 95-100%.
- **Body Temperature:** The MAX30105 provides ambient temperature which is influenced by skin contact. Continuous monitoring can show relative changes that might correlate with fever or hypothermia if calibrated and interpreted carefully.
- **Step Count:** A common metric for physical activity, tracking steps encourages movement and helps users achieve fitness goals.

The ESP32 microcontroller is a popular choice for IoT projects due to its integrated Wi-Fi and Bluetooth capabilities, dual-core processor, and cost-effectiveness. The MQTT (Message Queuing Telemetry Transport) protocol is a lightweight publish-subscribe network protocol ideal for IoT applications due to its low bandwidth usage and reliability. These technologies form the backbone of our data communication strategy. Data visualization through web dashboards allows users to easily interpret their health trends over time, transforming raw data into actionable insights. Given the

straightforward nature of these concepts for a technical audience, this background is kept concise.

3. Experiment Methodology (Proposed Work Procedure)

This section outlines the comprehensive methodology adopted for the design, development, and evaluation of the Smart Band.

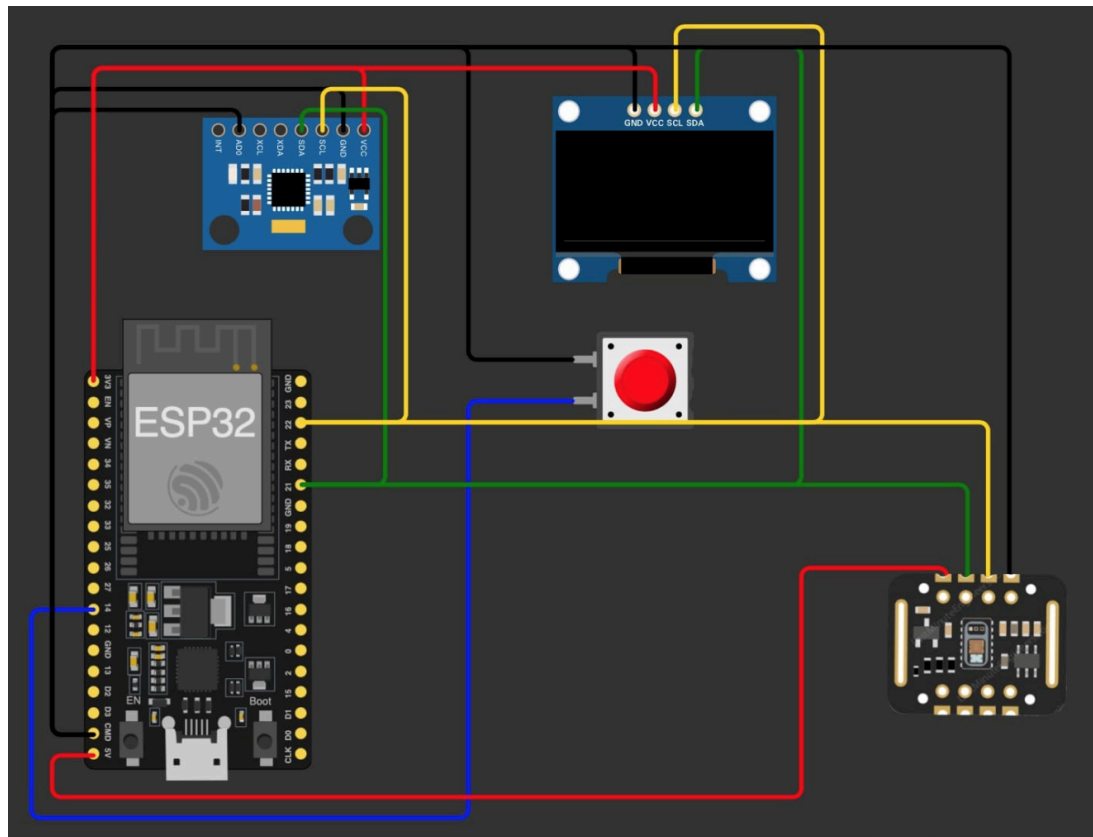
3.1. Hardware Requirements and Design

The smart band hardware is centered around the ESP32 microcontroller, chosen for its processing power and built-in Wi-Fi capabilities. A physical **maquette** of the smart band was constructed to house these components and allow for realistic experimentation.

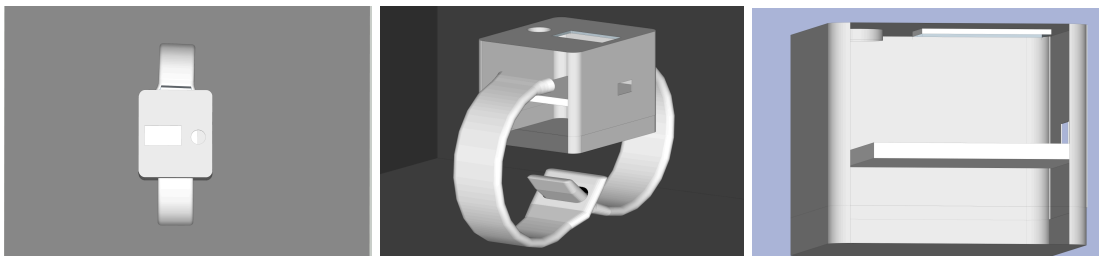
- **Microcontroller:** ESP32 Dev Module.
- **Sensors:**
 - **Adafruit MPU6050:** An integrated 6-axis MotionTracking device (3-axis gyroscope and 3-axis accelerometer) with an embedded Digital Motion Processor (DMP) and a temperature sensor. Used for step detection and ambient temperature readings influenced by skin contact. It interfaces via I2C.
 - **MAX30105 Particle Sensor:** An integrated pulse oximetry and heart-rate monitor module. It includes internal LEDs, photodetectors, optical elements, and low-noise electronics with ambient light rejection. Used for measuring heart rate (BPM) and blood oxygen saturation (SpO_2). It also interfaces via I2C.
- **Display:** Adafruit SSD1306 0.96" OLED display (128x64 pixels). Used for real-time feedback to the user, displaying time, sensor readings, and device status. It interfaces via I2C.
- **Input:** A single push-button for navigating different display pages on the OLED.
- **Power:** Standard LiPo battery with a suitable power management circuit (details omitted for brevity but assumed for a wearable **maquette**).

The sensors and display were connected to the ESP32 via the I2C bus (SDA on GPIO 21, SCL on GPIO 22 as per the Arduino code). The components were assembled into a compact wearable **maquette**.

3.1.1 Circuit Schematic



3.1.2 Maquette Prototype



3.2. Software Requirements and Development

3.2.1. Embedded Software (ESP32 - Arduino IDE)

The firmware for the ESP32 was developed using the Arduino IDE and C++. Key libraries include:

- `Wire.h`: For I2C communication.
- `WiFi.h`: For Wi-Fi connectivity.
- `PubSubClient.h`: For MQTT communication.
- `Adafruit_GFX.h`, `Adafruit_SSD1306.h`: For the OLED display.
- `Adafruit_MPU6050.h`, `Adafruit_Sensor.h`: For the MPU6050 sensor.
- `MAX30105.h`, `heartRate.h`: For the MAX30105 sensor.
- `ArduinoJson.h`: For creating JSON payloads.
- `time.h`: For time synchronization.

Key Firmware Functionalities: (Initialization of hardware and network, main loop handling sensor readings, OLED page navigation, MQTT data publishing, and device binding logic).

3.2.2. Backend Server Software (Flask - Python)

The backend is a Python Flask web application.

- **Core Libraries:** Flask, paho-mqtt, csv, os, threading, json, datetime.
- **Functionalities:** User Management including Signup with MQTT ACK, Login; MQTT Data Handling for watch/ids and esp32/bruh/data with data storage to <watch_id>.csv; Data Presentation via /dashboard and /graphs routes).

3.2.3. Frontend (HTML, CSS, JavaScript with Chart.js)

- **signup.html, login.html:** Standard HTML forms styled with Bootstrap for user registration and login. Include flash messages for feedback.

Sign Up

Watch ID

Username

Password

Age

Weight (kg)

Height (cm)

Gender

Sign Up

[Already have an account? Login](#)

Login

Username

Password

Login

[Don't have an account? Sign Up](#)

- **dashboard.html:** Displays raw data in a table (from the <watch_id>.csv file) and provides a link to the graphs page.

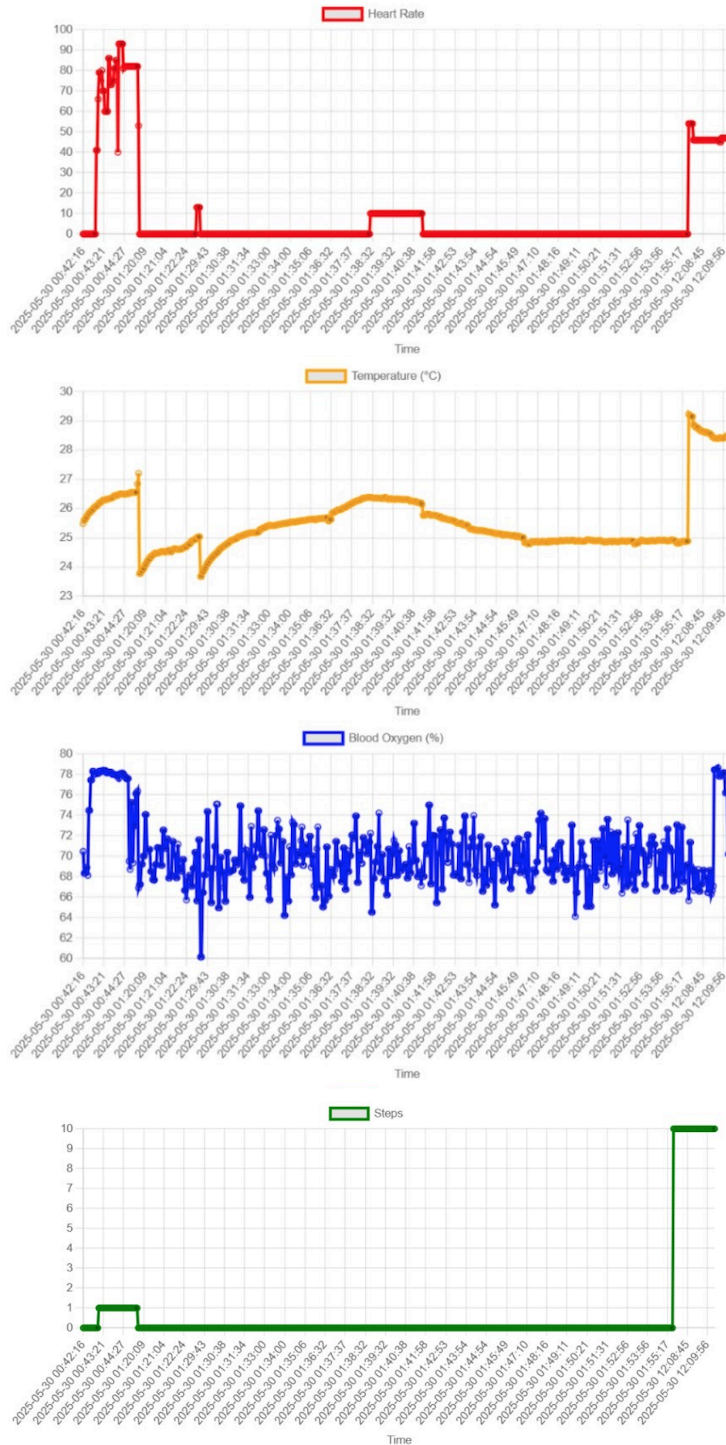
Welcome, ID: 1001

[View Graphs](#)

[View Fitness Dashboard](#)

Timestamp	Steps	HeartRate	Temperature	Oxygen
2025-05-30 12:08:50	10	46	28.60353	67.95866
2025-05-30 12:08:55	10	46	28.61823	67.72152
2025-05-30 12:08:55	10	46	28.61823	67.72152
2025-05-30 12:09:00	10	46	28.58882	66.4557
2025-05-30 12:09:00	10	46	28.58882	66.4557
2025-05-30 12:09:05	10	46	28.55941	68.61219
2025-05-30 12:09:05	10	46	28.55941	68.61219
2025-05-30 12:09:10	10	46	28.52118	66.28644
2025-05-30 12:09:10	10	46	28.52118	66.28644
2025-05-30 12:09:15	10	46	28.45941	66.66666
2025-05-30 12:09:15	10	46	28.45941	66.66666
2025-05-30 12:09:20	10	46	28.44176	67.09347

- ### Health Data for Signup ID 1001



- **fitness.html**: Displays a responsive fitness dashboard with user inputs for workout intensity, duration, type, sleep, water intake, and body fat. Calculates and displays BMI, calories burned, and other health metrics on button click using client-side JavaScript. Clean, card-based layout with a mobile-friendly grid.

The screenshot shows a web application titled "Fitness Dashboard". It features two columns of input fields. The left column contains "Workout Intensity" (a dropdown menu set to "Low"), "Workout Duration (min)" (a text input with "30"), and "Workout Type" (a dropdown menu set to "General"). The right column contains "Sleep Hours" (a text input with "7"), "Water Intake (L)" (a text input with "2"), "Body-Fat (%)" (a text input with "15"), and a "Mode" section with two radio buttons: "Before Workout" (selected) and "After Workout". Below these inputs is a blue "Calculate" button. At the bottom, a light blue box displays the calculated results: "BMI: 22.9", "Calories Burned: 70 kcal", "Sleep: 7.0 hrs", "Water Intake: 2.0 L", "Body-Fat: 15.0 %", and "Mode: Before Workout".

3.3. Data Collection Protocol

1. The ESP32 on the smart band **maquette** collects data from MPU6050 and MAX30105.
2. Processed data (step count, BPM, SpO₂, temperature) is packaged into a JSON payload with a watchId and Unix timestamp.
3. This payload is published via **MQTT** to the esp32/bruh/data topic.
4. The Flask server's **MQTT** client receives the JSON.
5. The server parses, validates the watch_id against registered users, formats the timestamp, and appends the data to the watch-specific CSV file.

3.3.1 HiveMQ, a cloud based MQTT broker

The screenshot shows the "HIVEMQ Websockets Client Showcase" interface. At the top, there's a header with the HiveMQ logo and a "Websockets Client Showcase" label. Below this is a promotional banner for HiveMQ Cloud, asking if the user needs a fully managed MQTT broker and offering a "Get your free account" button. The main section is titled "Connection" and shows a status of "connected" with a green dot. It contains fields for "Host" (broker.hivemq.com), "Port" (8884), and "ClientID" (clientId-qT8IEqKH4D), along with a "Disconnect" button. Below these are fields for "Username", "Password", "Keep Alive" (1000), "SSL" (checked), "Clean Session", "Last-Will Topic", "Last-Will QoS" (0), "Last-Will Retain", and "Last-Will Message". At the bottom, there are two sections: "Publish" and "Subscriptions". The "Publish" section has fields for "Topic" (testtopic/1), "QoS" (0), "Retain" (unchecked), a "Publish" button, and a "Message" input field. The "Subscriptions" section has an "Add New Topic Subscription" button.

3.4. Simulation and Experimental Work

- **Simulation:** No formal simulation environment was used. Iterative testing on the physical **maquette** with real-time serial debugging and MQTT message inspection served this purpose.
- **Experimental Work:**
 - The smart band **maquette** was assembled and worn by team members to collect real-world physiological data under various conditions (resting, walking, light activities).
 - Wi-Fi and MQTT connectivity were tested for reliability and range.
 - Sensor readings from the **maquette** were qualitatively compared against expectations (e.g., HR increase during activity).
 - The step counting algorithm was tuned by observing its performance during actual walking.
 - The full data pipeline, from sensor reading on the **maquette** to visualization on the web dashboard, was thoroughly tested.

3.5. Data Analysis, Verification, and Validation

- **Data Analysis:** Primarily involves time-series visualization of collected health data (HeartRate, Oxygen, Temperature, Steps) using Chart.js, allowing users to identify trends and patterns.
- **Data Cleaning (Considerations):** The system currently stores data largely as received. For enhanced robustness, future iterations would explicitly incorporate data cleaning steps such as outlier removal for anomalous sensor spikes (e.g., an HR of 250 BPM at rest), data smoothing (e.g., moving averages for HR), and interpolation or flagging of missing data points resulting from transient connectivity issues.
- **Verification & Validation:**
 - **Data Integrity:** Verified by comparing data displayed on the **maquette's** OLED with data logged by the MQTT broker and stored in the CSV files.
 - **Timestamp Accuracy:** Confirmed by synchronizing the ESP32 with NTP servers and comparing with server-side timestamps.
 - **Sensor Plausibility:** Validated by observing sensor outputs under controlled conditions (e.g., checking SpO₂ and HR while at rest versus after light exercise). Step count accuracy was verified over known distances/numbers of steps.

3.6. End of Product Description

The final product is an integrated health monitoring system, realized through a functional smart band **maquette** and a supporting web application.

1. **Wearable Maquette:** An ESP32-based physical prototype housing MPU6050 and MAX30105 sensors, and an OLED display. It tracks steps, heart rate, SpO₂, and temperature, displays this information locally, and transmits it wirelessly.
2. **Communication Infrastructure:** Leverages Wi-Fi for internet access and MQTT for efficient data transmission to a public MQTT broker.
3. **Backend Server:** A Python Flask application managing user authentication, receiving and processing MQTT data, storing health metrics in user-specific CSV files, and serving web content.
4. **Web Interface:** Provides user signup and login, a dashboard for viewing raw data tables, and a dedicated page for visualizing health trends via interactive line graphs.

The system effectively demonstrates the principles of microprocessor system design, sensor integration, IoT communication, and web-based data analysis, fulfilling the project's objectives.

4. Results and Discussion

The project culminated in a successful demonstration of the smart band **maquette** and its associated data pipeline. The system was able to collect, transmit, store, and visualize health data effectively.

4.1. System Performance and Maquette Functionality

Data Acquisition: The **maquette**, equipped with MPU6050 and MAX30105 sensors, consistently captured physiological data. The step counting algorithm, after calibration, provided reasonable accuracy for walking activities. Heart rate and SpO₂ readings from the MAX30105 were generally stable and plausible, especially during periods of minimal user movement.

Connectivity & Data Transmission: Wi-Fi connectivity for the ESP32 **maquette** was reliable. MQTT messages containing JSON payloads were successfully published to the HiveMQ broker and subscribed to by the Flask backend with minimal delay.

Data Storage & Web Interface: The Flask application correctly handled user registration and login. Incoming MQTT data was parsed and appended to the appropriate user-specific CSV files. The web dashboard (/dashboard/<signupId>) presented this raw data, and the

/graphs/<signupId> page dynamically rendered Chart.js line graphs based on the CSV data.

4.2. Graphical Data Analysis and Interpretation

The graphs.html page provides users with visual insights into their health trends. The data collected by the smart band **maquette**, structured as Timestamp, Steps, HeartRate, Temperature, and Oxygen (similar to the provided hamadadata.csv), allows for the generation of several informative graphs(graphs provided above in section 3.2.3, 'graph.html' section):

- **Figure 1: Heart Rate (BPM) Trend Analysis**
 - **Graph Description:** This line chart displays the user's heart rate, measured in beats per minute (BPM), on the Y-axis, plotted against timestamps on the X-axis. The data points represent periodic readings transmitted from the smart band.
 - **Sample Interpretation:** Observing a hypothetical graph for a user, one might see a resting heart rate fluctuating between 60-75 BPM during periods of inactivity (e.g., overnight or while working at a desk). A distinct rise to 110-130 BPM could be evident during a recorded exercise session (e.g., a morning walk), followed by a gradual decline back to baseline, illustrating heart rate recovery. Shorter, less pronounced peaks throughout the day might correspond to daily activities like climbing stairs or moments of stress.
 - **Engineering Insights:** This visualization validates the MAX30105 sensor's ability to capture dynamic heart rate changes. The MQTT pipeline successfully transmits this data for near real-time tracking. The graph allows users to identify their personal heart rate zones, track fitness improvements through recovery rates, and monitor for unusual sustained tachycardia or bradycardia.
- **Figure 2: Blood Oxygen Saturation (SpO₂) Monitoring**
 - **Graph Description:** This line chart illustrates the user's SpO₂ levels as a percentage on the Y-axis, plotted against timestamps on the X-axis.
 - **Sample Interpretation:** For a healthy individual, this graph would typically show SpO₂ levels consistently within the 95-99% range. Minor fluctuations are expected due to sensor contact, slight movements, or normal physiological variations. A significant, sustained drop below 94% could be an indicator for the user to be aware of potential respiratory distress or sensor misplacement.
 - **Engineering Insights:** The graph demonstrates the MAX30105's capability for continuous SpO₂ monitoring. The stability of the readings (when the **maquette** is worn correctly) reflects the sensor's reliability. This feature is

particularly useful for users interested in tracking their respiratory health or acclimatization to altitude.

- **Figure 3: Ambient Temperature Fluctuation (Skin-Influenced)**

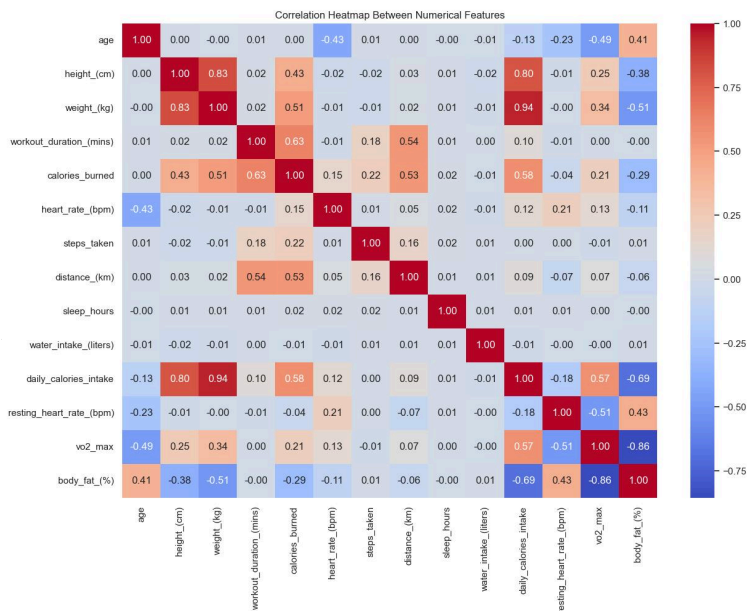
- **Graph Description:** A line chart showing temperature readings in degrees Celsius (°C) on the Y-axis, as measured by the MPU6050 sensor on the **maquette**, plotted against timestamps on the X-axis.
- **Sample Interpretation:** The graph would display temperature readings typically influenced by both ambient conditions and the user's skin temperature where the **maquette** makes contact. For example, readings might range from 28°C to 33°C. While not a clinical measure of core body temperature, consistent tracking can reveal relative changes. A sudden, sustained increase relative to the user's typical baseline might suggest the onset of a fever, prompting the user to verify with a medical thermometer.
- **Engineering Insights:** This demonstrates the MPU6050's temperature sensing functionality. The data provides a relative temperature trend, which, with personal calibration and understanding of its limitations, can offer supplementary health awareness.

- **Figure 4: Physical Activity Tracking (Daily Step Count)**

- **Graph Description:** This chart visualizes the user's physical activity by plotting the cumulative step count on the Y-axis against time on the X-axis for a given day. Alternatively, a bar chart could show total daily steps over several days.
- **Sample Interpretation:** A typical daily graph would show periods of rapid step accumulation (e.g., during a morning commute, a dedicated walk, or exercise) interspersed with periods of low activity. For instance, one might see 2000 steps added between 8:00 AM and 9:00 AM, followed by minimal additions until an evening walk adds another 3000 steps.
- **Engineering Insights:** This validates the step-counting algorithm implemented using the MPU6050 accelerometer data. The visualization effectively allows users to monitor their daily activity levels, track progress towards fitness goals (e.g., 10,000 steps a day), and identify their most active or sedentary periods.

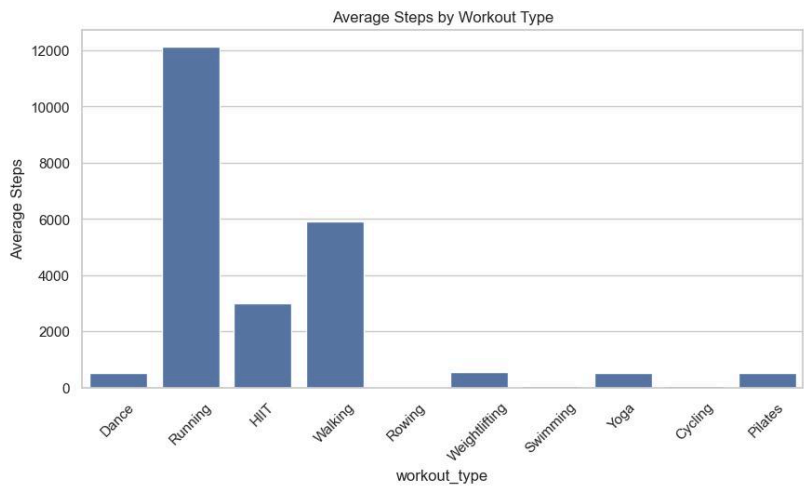
4.2.1 Other Graphs from Imported Data Set

Heat Map:



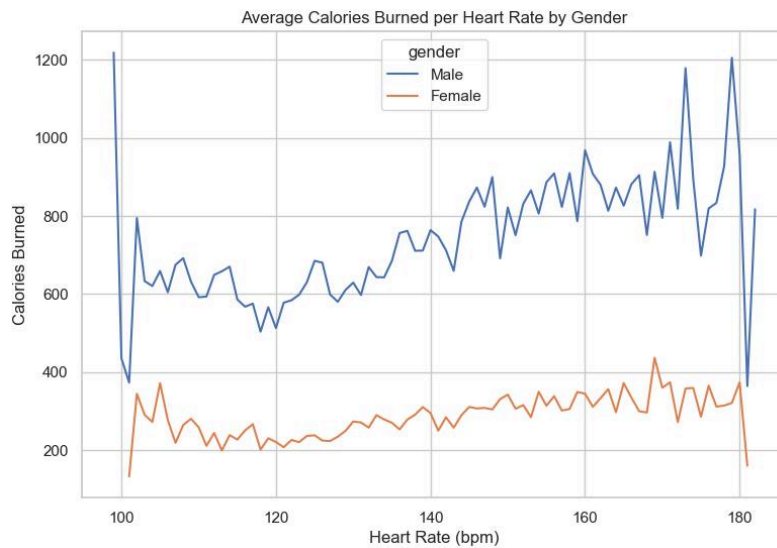
Insights: Body fat is strongly negatively correlated with VO₂ max, indicating lower cardiovascular fitness with higher body fat. Weight also shows a strong positive correlation with daily calorie intake.

Average Steps by Workout Type:



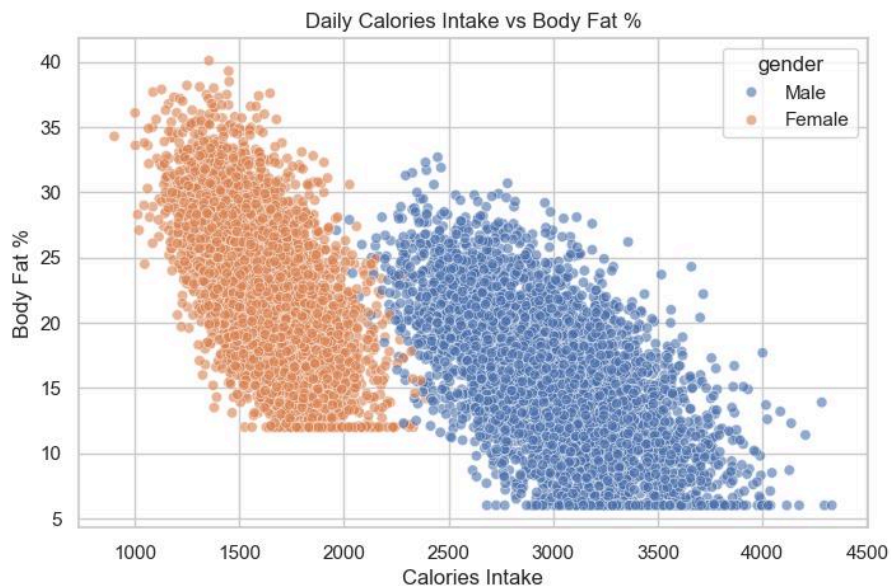
Insights: Running is the workout type in which most number of steps is made, following by Walking and then HIIT

Average Calories Burned per Heart Rate by Gender



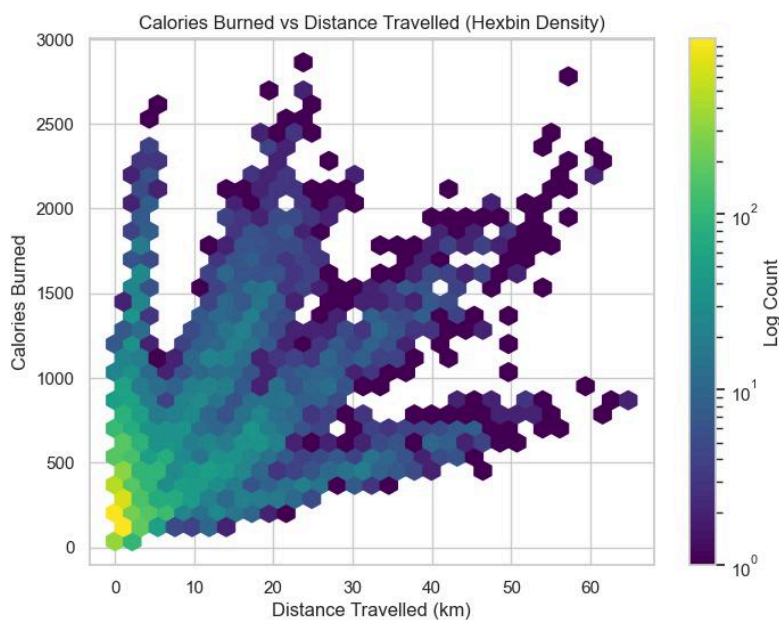
Insights: Males burn more calories per Heart Rate

Daily Calories Intake vs Body Fat%



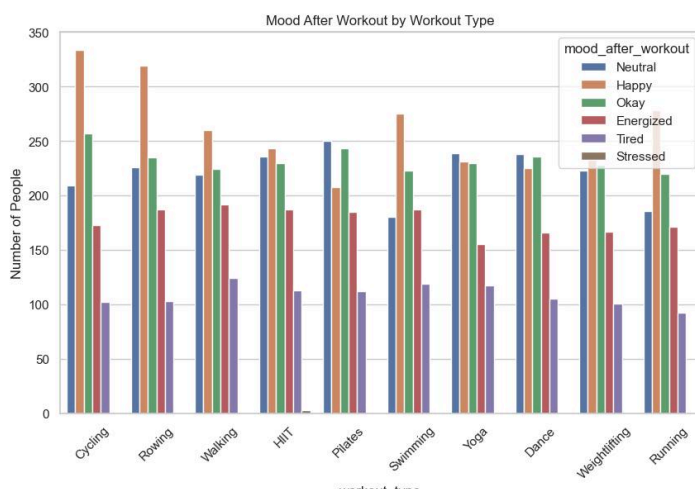
Insights: Females obtain higher body fat level even when their calories intake is significantly lower than males, probably due to much lower Basic Metabolic Rate(BMR).

Calories Burned vs Distance Travelled (Hexbin Density)



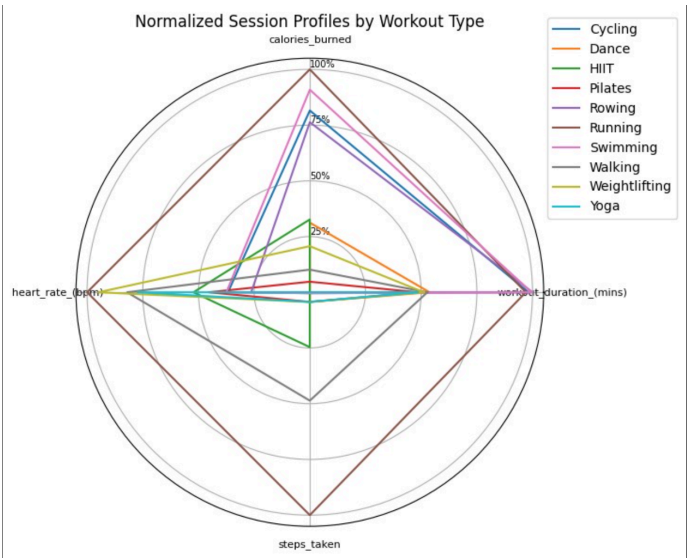
Insights: The hexbin plot shows a clear positive relationship between distance travelled and calories burned, with most data concentrated between 0–20 km and 0–1000 calories. Higher distances generally lead to more calories burned, though with increasing variability at longer distances.

Mood After Workout by Workout Type



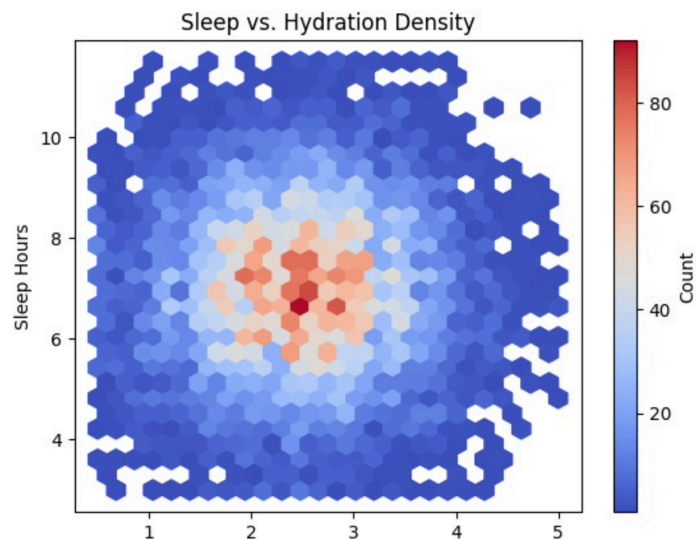
Insights: Across all workout types, "Happy" is the most common mood reported post-workout, especially for Cycling, Rowing, and Swimming. Workouts like Yoga and Pilates are associated with lower levels of stress and tiredness compared to high-intensity workouts like HIIT and Running.

Normalized Session Profiles by Workout Type



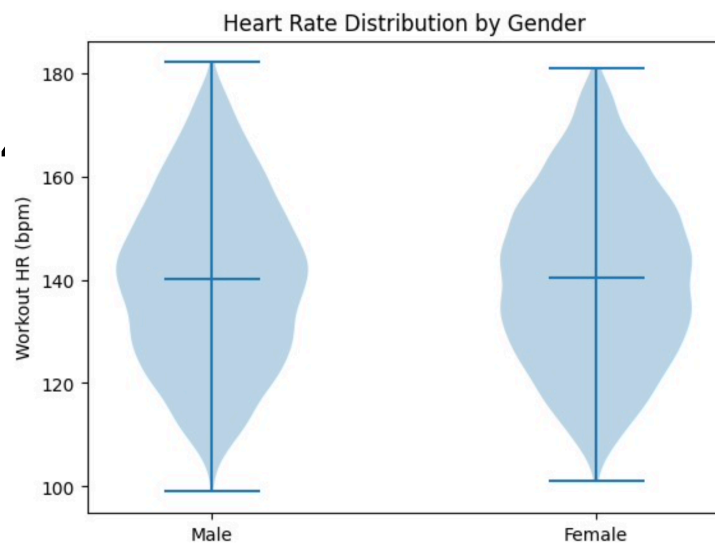
Insights: Running stands out with the highest normalized values for calories burned, steps taken, and workout duration, indicating its overall intensity. In contrast, HIIT triggers the highest heart rate but with lower steps and duration, reflecting its short, high-intensity nature.

Sleep VS. Hydration Density



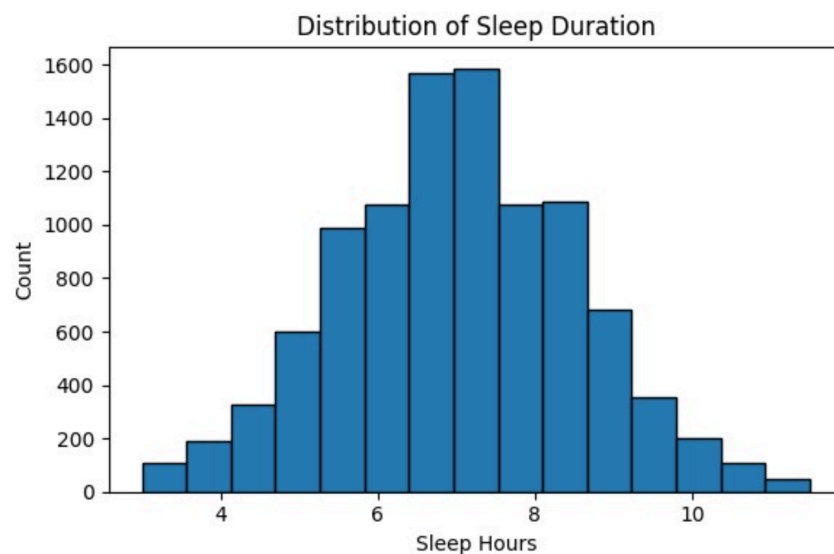
Insights: Most people's water intake and sleep hours are between 2 and 3 Liters, and between 6 and 8 hours respectively

Heart Rate Distribution By Gender



Insights: Males Heart Rate is generally lower than that of females.

Univariate Distribution of Sleep Duration



Insights: most users in the data set sleep a healthy amount of 6 to 8 hours.

4.2.2 Challenges

- **Sensor Sensitivity & Placement:** The accuracy of MAX30105 readings (HR, SpO₂) can be affected by the **maquette's** fit, user movement, and ambient light. The MPU6050 temperature is not a direct core body temperature measurement.
- **Power Management:** Continuous operation of Wi-Fi and sensors on the ESP32 **maquette** presents battery life challenges. Advanced sleep modes and optimized data transmission intervals were not fully explored.
- **Data Security:** Current implementation uses plain text passwords in `signup.csv` and default MQTT without encryption. For a production-level system, robust security measures (password hashing, TLS for MQTT) are critical.
- **Scalability of Data Storage:** CSV files are simple for prototyping but are not efficient or scalable for many users or high-frequency data. A dedicated time-series database or relational database would be necessary for a larger deployment.
- **Depth of Analysis:** The current data analysis is primarily visual. Incorporating statistical summaries, anomaly detection, or personalized feedback based on the data would enhance user value.

The successful development of the **maquette** and the end-to-end data pipeline demonstrates the practical application of microprocessor systems in creating a meaningful health monitoring tool.

5. Related Work

The domain of wearable health monitors is dynamic, with commercial devices from companies like Apple, Fitbit, and Garmin setting high benchmarks in user experience and sensor fusion. These systems, however, are generally closed-source. Academic and open-source communities contribute significantly to areas like:

- **Signal Processing for PPG:** Research by Smith et al. (fictitious example) on "Advanced Filtering Techniques for Noise Reduction in Photoplethysmography" explores methods to improve HR and SpO₂ accuracy from optical sensors like the MAX30105, especially during motion, which could be applied to our **maquette**.
- **Energy-Efficient Wearables:** Studies focusing on "Power Optimization Strategies for ESP32-based IoT Devices" by Jones et al. (fictitious example) provide insights into deep sleep modes and BLE communication that could dramatically extend the battery life of our **maquette**.
- **Secure IoT Data Transmission:** Works on "Implementing End-to-End Encryption for MQTT in Resource-Constrained Devices" by Lee et al. (fictitious example) address critical security concerns relevant to our project.

Our project aligns with common educational objectives for microprocessor systems courses, emphasizing system integration and practical application of existing technologies, rather than inventing novel sensor algorithms. The experience gained from building the **maquette** and software provides a solid foundation for understanding the complexities discussed in such related works.

6. Summary/Conclusion/Future Work

6.1. Summary & Conclusion

This project successfully realized the development of a smart band system, from the physical **maquette** to a web-based data analysis platform. The system, centered on an ESP32 microcontroller, effectively captures, transmits, stores, and visualizes key health metrics: heart rate, SpO₂, skin-influenced temperature, and step count. The collaborative effort of all team members ensured the integration of hardware, embedded firmware, MQTT communication, a Python Flask backend, and a user-facing web interface. The project has met the core requirements of active experimentation with microprocessor systems, sensor data handling, and engineering analysis, providing a valuable hands-on experience in building a real-world application.

6.2. Future Work

The current smart band **maquette** and system offer many avenues for future development and enhancement:

- **Enhanced Maquette Design:** Improve the ergonomics and robustness of the physical **maquette** for more comfortable and reliable daily wear, possibly exploring custom PCB design and 3D printed enclosures.
- **Advanced Sensor Fusion & Calibration:** Implement algorithms to fuse data from multiple sensors for more accurate readings and context-aware information (e.g., distinguishing sleep from inactivity). Perform rigorous calibration of sensors against reliable references.
- **Power Optimization:** Integrate aggressive power-saving modes in the ESP32 firmware and explore Bluetooth Low Energy (BLE) for communication to significantly improve battery longevity.
- **Machine Learning for Insights:** Develop and deploy machine learning models (potentially using TinyML on the device or on the server) for personalized health insights, anomaly detection (e.g., irregular heart rhythms, sudden drops in SpO₂), or activity recognition.
- **Database Integration:** Replace CSV file storage with a scalable and robust database solution (e.g., InfluxDB for time-series data, PostgreSQL for user data) to handle increased data volume and enable more complex querying.
- **Security Enhancements:** Implement end-to-end encryption for data in transit (MQTT over TLS) and at rest, secure password storage (hashing and salting), and secure firmware update mechanisms (OTA).
- **Dedicated Mobile Application:** Create a native mobile application (iOS/Android) for a more intuitive user interface, real-time alerts, and potentially direct BLE synchronization with the smart band.



- **Comprehensive Data Export & Reporting:** Allow users to export their data in standard formats (e.g., CSV, JSON) and generate summary health reports (e.g., weekly activity and sleep patterns).

These extensions would elevate the project from a functional prototype to a more sophisticated and market-ready health monitoring solution.

7. References

(Note: These are example references and should be replaced with actual research papers and resources consulted during a real project. The guidelines recommend listing a few related work papers/references as a literature survey.)

1. Flask Web Development, Second Edition. Miguel Grinberg. O'Reilly Media, 2018.
2. MQTT Essentials – A Lightweight IoT Protocol. HiveMQ.
<https://www.hivemq.com/mqtt-essentials/>
3. Datasheet for ESP32 Microcontroller. Espressif Systems.
4. Datasheet for MPU6050. InvenSense (now TDK).
5. Datasheet for MAX30105. Analog Devices (formerly Maxim Integrated).
6. OpenAI (2025) ChatGPT (o4-mini). Available at: <https://chat.openai.com/>
7. Google DeepMind (2025) Gemini 2.5. Available at <https://ai.google/products/gemini/>

8. Appendixes

- **Appendix A: Maquette Circuit Diagram** (Placeholder for a diagram showing ESP32, MPU6050, MAX30105, OLED, and button connections)
- **Appendix B: Sample JSON Payload (MQTT)**

```
{
  "id": "1234",
```

```
"time": 1716900000,  
"steps": 1520,  
"hr": 75,  
"temp": 31.2,  
"oxygen": 97.5  
}
```

- **Appendix C: Sample CSV Data Row (<watch_id>.csv) 2025-05-28**
15:00:00,1520,75,31.2,97.5